

ORIGINAL ARTICLE

ERIC-PCR-based molecular typing of multidrug-resistant *Escherichia coli* isolated from houseflies (*Musca domestica*) in the environment of milk and meat shopsC. Chandrakar¹ , S. Shakya¹, A. Patyal¹, A. Jain², S.L. Ali³ and O.P. Mishra⁴¹ Department of Veterinary Public Health and Epidemiology, Dau Shri Vasudev Chandrakar Kamdhenu Vishwavidyalaya, Durg, India² Department of Animal Genetics and Breeding, Dau Shri Vasudev Chandrakar Kamdhenu Vishwavidyalaya, Durg, India³ Department of Veterinary Medicine, Dau Shri Vasudev Chandrakar Kamdhenu Vishwavidyalaya, Durg, India⁴ Department of Veterinary Physiology and Biochemistry, Dau Shri Vasudev Chandrakar Kamdhenu Vishwavidyalaya, Durg, India

Significance and impact of the study: Houseflies (*Musca domestica*) are endophilic cosmopolitan pests of medical and veterinary importance. These are associated with regions of human activities such as food centres, butcher houses, animal farms and surrounding areas and create an annoyance to people, poultry and domestic farm animals. The present study reports that housefly samples collected from milk and meat outlets were positive for multidrug-resistant (MDR) *E. coli* and harbour *bla*_{CTX-M} and *bla*_{TEM} genes in plasmid along with other resistant genes for tetracyclines and quinolones. The presence of MDR genes in bacteria isolated from housefly samples is of serious concern from public health and food safety points of view. Investigation of genetic determinants of MDR in bacteria would be of great value for formulating and validating suitable control strategies to combat the spread of antimicrobial resistance.

Keywordshousefly, milk and meat shops, molecular typing, multidrug-resistant, vector, *E. coli*.**Correspondence**

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2022/LAMICRO-2022-0270.R2: received 13 May 2022, revised 22 August 2022 and accepted 23 August 2022

doi:10.1111/lam.13821

Abstract

The emergence and spread of antimicrobial resistance have become a major global public health concern. A component of this problem is the spread of antibiotic-resistant bacteria. Flies move freely between habitats of food-producing animals and human beings and thus have great potential for dissemination of antimicrobial-resistant bacteria from a contaminated environment to milk and meat markets, posing potential hazards for consumers. During the present study, a total of 150 houseflies were captured from milk and meat shops located in Durg and Raipur city of Chhattisgarh, India. The *Escherichia coli* were isolated from houseflies and characterized on the basis of cultural and molecular tests. Further, the isolates were subjected to antimicrobial susceptibility testing against frequently used antibiotics using the disk diffusion method. The antibiotic resistance genes and *int1* gene were detected using polymerase chain reaction (PCR). A total of 45 *E. coli* isolates were obtained from the fly samples with an overall prevalence rate of 30.0%. Antibioqram results confirmed that *E. coli* isolates were resistant to multiple antibiotics. Out of the (45) isolates of *E. coli*, 17 (37.8%) isolates were extended-spectrum beta-lactamase (ESBL) producer and multi-drug-resistant (MDR). Out of the ESBL and MDR *E. coli* isolates, *bla*_{CTX-M} (24.4%), *bla*_{TEM} (11.1%), *tetA* (28.8%), *tetB* (26.7%), *gyrA* (26.7%), *parC* (31.1%) and *int1* genes (15.5%) were detected but none of the isolates were found positive for *bla*_{SHV} gene. Findings of the present study confirm that MDR *E. coli* are widely distributed in houseflies and play an important role in the transmission of antibiotic-resistant bacteria from contaminated environments to milk and meat shop environment.

Introduction

The increasing trend of multi-drug-resistant (MDR) has become a global issue affecting the important role of antimicrobial therapies. The World Health Organization (WHO) has identified antibiotic-resistant pathogens as one of the biggest challenges and threats to human health (WHO 2014). Antimicrobial resistance (AMR) is estimated to cause 700 000 deaths each year across the world (Willyard 2017). The impact of poor food hygiene on public health due to an increased incidence of gastrointestinal diseases is even further aggravated by the presence of flies that serve as vectors for a variety of pathogenic microorganisms. Flies can transmit MDR microorganisms by mechanical translocation, regurgitation (bio-enhanced transmission) and defecation (Kobayashi *et al.* 1999). MDR bacteria may become a part of the gut flora of flies thus carrying the bacteria for their life span and contaminating their environment via faeces and/or regurgitation. *Escherichia coli*, a commensal bacterium of humans and animals, is a significant cause of gastrointestinal disorders (Ryan and Ray 2004). The presence of drug-resistant genes in *E. coli* is one of the most frequently reported complications during the management of enteric diseases (Nakhjavani *et al.* 2013). Further, *E. coli* is sometimes used as a sentinel for monitoring AMR in faecal bacteria due to its frequent and persistent presence in a wide range of hosts (Erb *et al.* 2007). In Chhattisgarh, most of the meat and milk shops are not fly protected which poses a risk of transmission of AMR bacteria through flies to consumers. Studies carried out in various regions have indicated the presence and transmission of MDR genes but so far no information in this regard is available from the state of Chhattisgarh. Therefore, the present study was conducted to investigate whether flies persistent in the market areas carry antibiotic-resistant *E. coli* that may be of serious threat to public health.

Results and discussion

Prevalence of *E. coli* in fly samples

During the study, a total of 150 fly samples were collected and screened for *E. coli*. On microbiological and molecular analyses, a total of 27 (37.06%) isolates from meat shops ($n = 77$) and 18 (24.65%) isolates from milk outlets ($n = 73$) were confirmed as *E. coli*. The result obtained in the present study is in concordance with the positivity (36.5%) reported in Iranian houseflies (Vazirianzadeh *et al.* 2008). However, a significantly lower prevalence of 6.7% was reported in house fly samples collected from Vellore city, India (Collinet-Adler *et al.* 2015). Higher prevalence rates ranging between 51.4% and 77% were reported

from other geographic regions (Barreiro *et al.* 2013; Liu *et al.* 2013; Nazari *et al.* 2017). In the present study, houseflies were collected from meat shop environments, where live birds are slaughtered and their slaughter waste is not disposed of hygienically. This poorly managed waste is not only associated with a high prevalence of *E. coli* but also attracts flies. Several studies have reported that flies from such environments harbour various pathogens (De Jesus *et al.* 2004; Schaumburg *et al.* 2016; Cervelin *et al.* 2018). These *E. coli* harbouring flies may contaminate the meat-processing environment and other food products and cause food-borne diseases. Flies could easily transmit the pathogenic *E. coli* to other distant places through the body surface (Songe *et al.* 2017).

Antimicrobial-resistant phenotypes of *E. coli*

In the current study, the antimicrobial susceptibility tests demonstrated that all 45 *E. coli* isolates from fly samples of milk and meat shops showed phenotypic resistance against all the 10 tested antimicrobials. The *E. coli* isolates were highly resistant to ampicillin (73.3%), cefotaxime (66.6%), oxytetracycline (66.6%) and cefixime (53.3%). The resistance rates of the *E. coli* isolates to other antimicrobials were between 37.7 and 48.8%. For the tested four classes of veterinary drugs, the *E. coli* isolates showed higher susceptibility to cefalexin, ampicillin-cloxacillin, and ciprofloxacin and had lower resistance rates of 37.7, 42.2 and 44.4%, respectively. *Escherichia coli* isolates from milk shop fly samples showed high resistance rates to ampicillin, cefotaxime, oxytetracycline and aztreonam, whereas in the case of meat shop fly, *E. coli* isolates of higher resistance rates were recorded for ampicillin, cefotaxime and oxytetracycline. *Escherichia coli* isolates from the meat shop were comparatively more resistant than isolates from the milk shop. Among 45 *E. coli* isolates, 44 (97.7%) isolates were resistant to two or more than two antibiotics. The highest multiple antibiotic resistance (MAR) index of 1.0 was observed in 4 (8.8%) isolates, followed by 0.9 in 1 (2.2%), 0.8 in 1 (2.2%), 0.7 in 7 (15.5%), 0.6 in 5 (11.1%), 0.5 in 7 (15.5%), 0.4 in 9 (20%), 0.3 in 8 (17.7%) and 0.2 in 2 (4.4%) isolates. MAR index values of >0.2 are regarded as an indicator of anthropogenic contamination, and values of <0.2 indicate that the environment of isolation has no history or very low use of antibiotics (Krumperman 1983).

Hierarchical clustering analysis allowed the segregation of the 45 *E. coli* isolates according to their phenotypic antibiotic resistance profiles (Fig. 1). Interestingly, separation was not observed among *E. coli* isolated from milk and meat shop environments and sampling sites. Overall, five main clusters and three singletons were identified. Cluster I included 8 *E. coli* isolates which showed

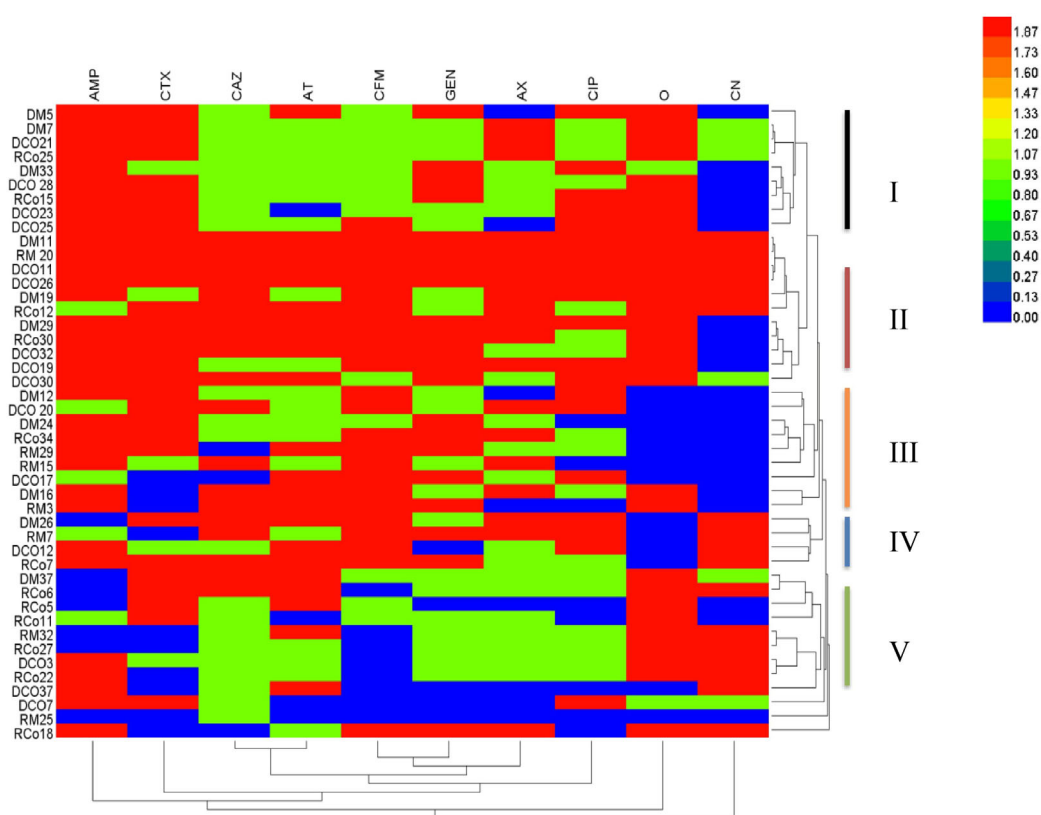


Figure 1 Two-way hierarchical clustering with a heat map of antimicrobial susceptibility profiles to 10 antibiotics of *E. coli* isolated in this study. The heatmap indicates the categorical value of susceptibility from each antibiotic as indicated by a color bar (blue = sensitive, green = intermediate, and red = resistant).

resistance to ampicillin, cefotaxime and oxytetracycline antibiotics. Cluster II was the largest and contained 12 *E. coli* isolates, out of which DM11, RM20, DCo11, and DCo21 demonstrated resistance to all tested antibiotics. This cluster also grouped more resistant isolates than the other groups. Cluster III consisted of 9 isolates, whereas clusters IV and V comprised 4 and 9 *E. coli* isolates, respectively. The majority of MDR *E. coli* were grouped in cluster II and cluster IV, whereas cluster III and cluster V were mainly comprised *E. coli* isolates resistant to two to four antibiotics (Fig. 1). MDR *E. coli* isolates grouped in different clusters had their origin in both the milk and meat shop environments, indicating that the houseflies shared a common environment of milk and meat shops.

All the *E. coli* isolates were further screened phenotypically for ESBL production. Among 45 *E. coli* isolates, 4 (22.22%) isolates from milk outlets and 13 (48.14%) isolates from meat shops were found positive for ESBL production (Fig. 2). These ESBL-producing isolates were also resistant to ciprofloxacin, oxytetracycline, gentamicin and identified as MDR. Antibiotic resistance in recent years has become an increasingly serious public health concern around the world (Xu et al. 2014). In

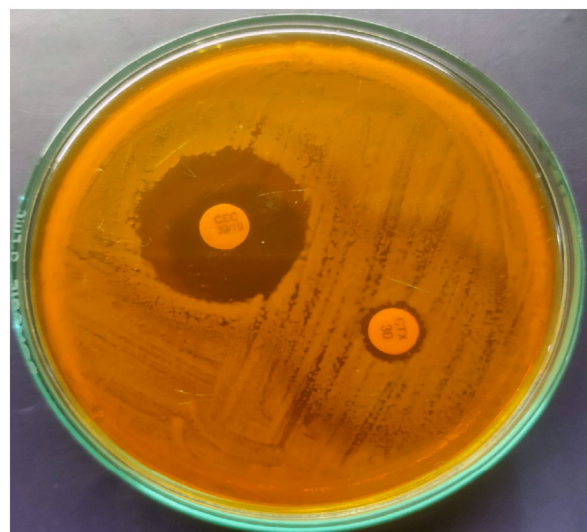


Figure 2 *Escherichia coli* isolate of fly sample showing phenotypic confirmatory test for ESBL production.

the present study, resistance against ampicillin (73.3%), cefotaxime (66.6%), oxytetracycline (66.6%) and cefixime (53.3%) were reported and these antibiotics were

found to have the lowest efficacy against *E. coli* recovered from the flies. *Escherichia coli* isolates were found to be more sensitive against cefalexin, ampicillin-cloxacillin, ciprofloxacin and gentamicin. Sobur et al. (2019) reported a similar finding from Bangladesh, where most of the fly recovered *E. coli* isolates were resistant to tetracycline (93.35%), erythromycin (91.8%) and ampicillin (61.75%). However, these *E. coli* isolates were more sensitive to gentamicin, ciprofloxacin, imipenem, colistin and chloramphenicol. The difference in resistant patterns could be attributed to antibiotic use patterns in the two countries as well as fly sampling locations. The *E. coli* isolated from fly samples of fish food markets in Zambia were also found resistant to ampicillin (100%), cefotaxime (100%), ceftazidime (95.2%), tetracycline (91%), gentamicin (91%) and ciprofloxacin (95.2%) (Songe et al. 2017). There are several reports on the isolation of pathogenic bacteria including MDR strains from the houseflies (Bouamama et al. 2010; Rahuma et al. 2005).

Antimicrobial-resistant genotypes of *E. coli* isolates

Out of the 45 *E. coli* isolates, 17 isolates presumptively positive for ESBL production and MDR were further subjected to the detection of seven antimicrobial resistance and *int1* genes (Table 1). The *parC* gene ($n = 14$; 31.1%) was the most frequently detected resistance gene, followed by *tetA* ($n = 13$; 28.9%), *gyrA* ($n = 12$; 26.7%), *tetB* ($n = 12$; 26.7%), *bla_{CTX-M}* ($n = 11$; 24.4%) and *bla_{TEM}* ($n = 5$; 11.1%). The detection rate for the *int1* gene was 15.5% ($n = 7$; Fig. 3). Although the *E. coli* isolates

showed high phenotypic resistance against the tested antimicrobials, the *bla_{SHV}* gene was not detected in any of the isolates. The most frequently detected resistance genes in *E. coli* isolates were five and four genes ($n = 10$; 22.2%), followed by six genes ($n = 3$; 6.6%), three genes ($n = 3$; 6.6%) and two genes ($n = 1$; 2.2%). Additionally, all studied genes, except *parC* and *gyrA*, were also detected in the plasmid DNA of isolated *E. coli*. Many isolates harbour resistant genes along with integrons on a plasmid. The plasmids have a reservoir of antibiotic-resistant genes and may transfer resistance genes horizontally from resistant bacteria to susceptible ones and thus play an important role in the adaptation of bacteria (Labate et al. 2009). Class I integrons on plasmids facilitate the conjugative-mediated transfer of AMR genes through site-specific recombination reaction mediated by the integron integrase (Deng et al. 2015). The presence of antibiotic resistance and *int1* genes in *E. coli* clearly indicates that houseflies play an important role in the dissemination of AMR genes, which is consistent with the previous report on AMR gene transfer within bacterial populations in flies (Liu et al. 2013). The present study is the first attempt to detect antibiotic-resistant genes in *E. coli* associated with houseflies of milk and meat outlets in Chhattisgarh, India.

Enterobacterial repetitive intergenic consensus-PCR typing

The determination of genomic diversity of 17 isolates of phenotypic ESBL producer and MDR *E. coli* was carried out using the ERIC-PCR typing method. The

Table 1 *Escherichia coli* isolates and their antimicrobial resistance profile

<i>Escherichia coli</i> Isolate code	Resistance to antimicrobials	Resistance genes
DC _O 11	AMP, CTX, CAZ, CFM, AXE, GEN, AT, CIP, O, CN	<i>bla_{CTX-M}</i> , <i>GyrA</i> , <i>parC</i> , <i>tetB</i> , <i>int1</i>
DC _O 25	AMP, CTX, CFM, CIP, O	<i>GyrA</i> , <i>parC</i> , <i>tetA</i> , <i>tetB</i> , <i>int1</i>
RM 20	AMP, CTX, CAZ, CFM, AXE, GEN, AT, CIP, O, CN	<i>bla_{CTX-M}</i> , <i>GyrA</i> , <i>parC</i> , <i>tetA</i> , <i>tetB</i>
DC _O 19	AMP, CTX, AXE, GEN, CIP, O	<i>bla_{TEM}</i> , <i>GyrA</i> , <i>parC</i> , <i>tetA</i> , <i>tetB</i> , <i>int1</i>
DC _O 23	AMP, CTX, CIP, O	<i>bla_{CTX-M}</i> , <i>GyrA</i> , <i>parC</i> , <i>tetA</i> , <i>tetB</i>
DM19	AMP, CAZ, CFM, AXE, CIP, O, CN	<i>bla_{CTX-M}</i> , <i>bla_{TEM}</i> , <i>GyrA</i> , <i>parC</i> , <i>tetA</i> , <i>tetB</i>
RC _O 7	AMP, CTX, CAZ, CFM, CIP (I), O (I)	<i>bla_{CTX-M}</i> , <i>GyrA</i> , <i>parC</i>
DC _O 21	AMP, CTX, AXE, CIP (I), O	<i>bla_{CTX-M}</i> , <i>GyrA</i> , <i>parC</i> , <i>tetA</i>
RC _O 6	CTX, CAZ, AT, CIP (I), O	<i>GyrA</i> , <i>parC</i> , <i>tetA</i> , <i>tetB</i>
RC _O 12	CTX, CAZ, CFM, AXE, AT, CIP (I), O	<i>bla_{TEM}</i> , <i>tetA</i> , <i>tetB</i> , <i>int1</i>
RC _O 15	AMP, CTX, GEN, CIP, O	<i>bla_{CTX-M}</i> , <i>GyrA</i> , <i>int1</i>
DC _O 28	AMP, CTX, GEN, CIP (I), O	<i>parC</i> , <i>tetB</i>
DC _O 30	AMP, CTX, CAZ, GEN, AT, CIP, O	<i>bla_{CTX-M}</i> , <i>bla_{TEM}</i> , <i>parC</i> , <i>tetA</i> , <i>tetB</i>
DM29	AMP, CTX, CAZ, CFM, AXE, GEN, AT, CIP, O	<i>bla_{TEM}</i> , <i>GyrA</i> , <i>parC</i> , <i>tetA</i> , <i>tetB</i> , <i>int1</i>
RC _O 30	AMP, CTX, CAZ, CFM, AXE, GEN, AT, CIP (I), O	<i>bla_{CTX-M}</i> , <i>parC</i> , <i>tetA</i> , <i>int1</i>
DC _O 32	AMP, CTX, CAZ, CFM, GEN, AT, CIP (I), O	<i>bla_{CTX-M}</i> , <i>tetA</i> , <i>tetB</i>
RM11	AMP, CTX, CAZ, CFM, GEN, AT, CIP, O, CN	<i>bla_{CTX-M}</i> , <i>GyrA</i> , <i>parC</i> , <i>tetA</i>

D = Durg, R = Raipur, M = Milk shops, Co = Meat shops (Chicken, Fish and Chevron shops).

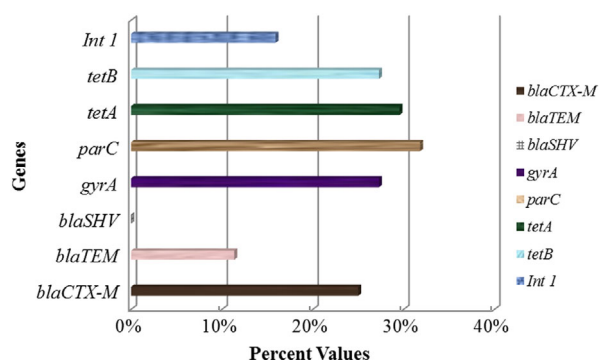


Figure 3 Prevalence of resistant genes shown by *E. coli* isolates from house fly samples.

electrophoretic analysis of the reaction products revealed multiple typing patterns in all isolates. The number of bands in particular electrophoretic paths ranged between 3 and 15 with product sizes up to 4000 bp. The analysis of ERIC-PCR profiles categorized 17 *E. coli* isolates into 15 ERIC types and 4 clusters (Figs 4 and 5). The results further revealed that ERIC types of all isolates were different except for E02 and E05 types.

In this study, ERIC-PCR typing showed 15 patterns for 17 *E. coli* isolates. Thirteen (76.47%) of these isolates displayed a separate ERIC profile. The uniqueness of ERIC patterns suggests that they had nonclonal transmission. However, four (23.52%) isolates had identical patterns, indicating a similar origin of dissemination. Overall, genetic analysis revealed that the majority of ESBL and MDR isolates were clonally unrelated, suggesting that the dissemination of these isolates was not from the same clonal group. Among several PCR-based methods, ERIC-PCR is simple, fast and affordable to differentiate types of strains and less dependent on variable bacterial growth factors (Ranjbar and Ghazi 2013; Ranjbar et al. 2014). ERIC-PCR is repeated sequences with 122–127 bps found in the Enterobacteriaceae family. These sequences and differences in their frequency of repetition are used as a molecular tool for genetic screening (Sharp and Leach 1996; Sedighi et al. 2020).

In the present study, ESBL and MDR *E. coli* were detected in flies captured from various milk and meat shop environmental settings. Houseflies are an important vector for the transmission of MDR bacteria (Yin

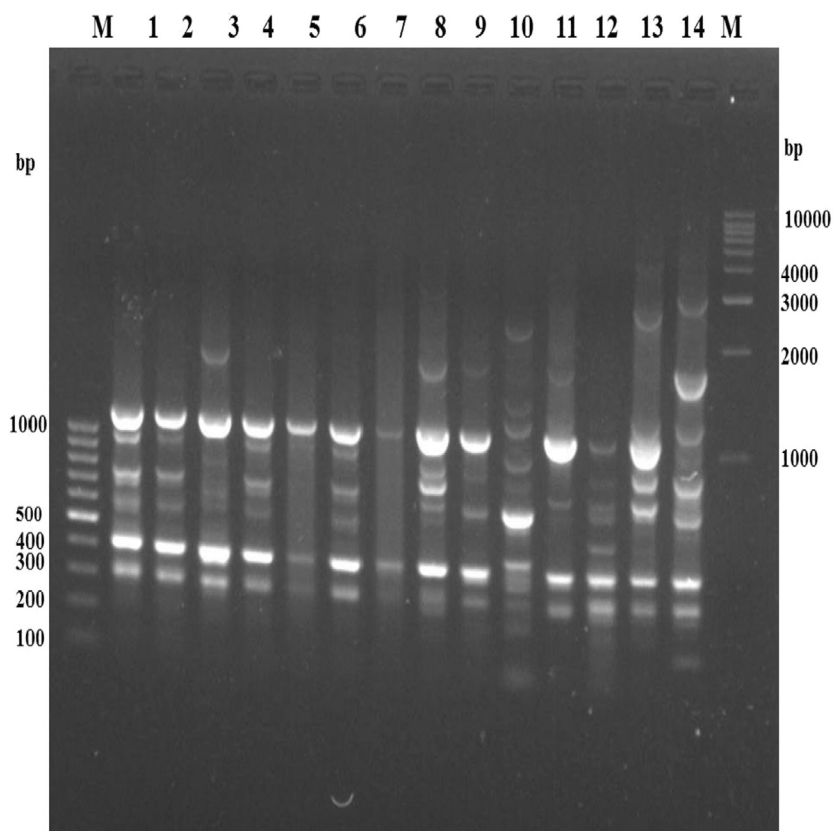


Figure 4 ERIC-PCR typing of *Escherichia coli* isolates from fly sample. lanes 1–14 ERIC profile of different *E. coli* isolates; Lane M –100 bp and 1 kb DNA ladder.

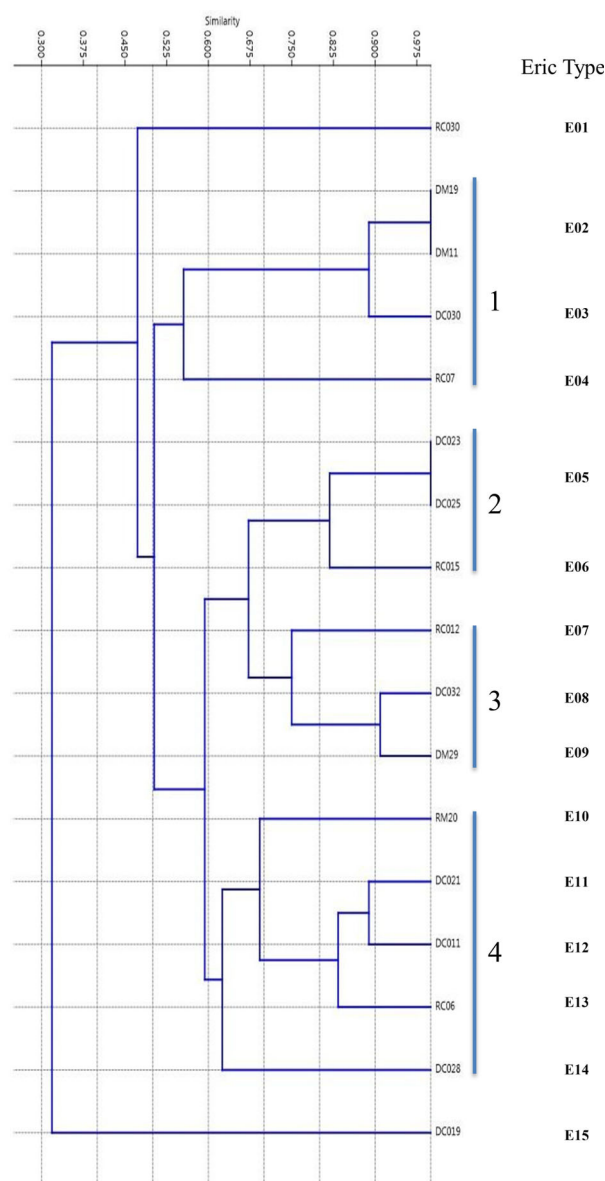


Figure 5 ERIC-PCR typing of *Escherichia coli* isolated from fly samples collected from milk and meat shops.

et al. 2022). This represents an insurmountable threat to consumer health. The findings of this study are of public health significance as a great proportion of humans are in contact with food corner and dairy farm/shop environments and may be exposed to infection with such MDR *E. coli*. Implementation of proper and strict sanitary measures within these environments and their surroundings is critical to reduce the threats of transmission of AMR genes through flies. This study generates baseline data on the role of houseflies in the dissemination of AMR bacteria, for the AMR National Action Plan of the Government of India.

Materials and methods

Sample collection and area of study

The sample size was estimated at a 95% confidence interval, using the online statistical calculator, Statulator (Dhand and Khatkar 2014). The prevalence of *E. coli* was hypothesized to be $4.0 \pm 5\%$ as per prevalence studies conducted on houseflies across the globe (Liu *et al.* 2013; Collinet-Adler *et al.* 2015; Parvez *et al.* 2016; Schaumburg *et al.* 2016). A total of 150 houseflies (*Musca domestica*) samples were collected from milk outlets ($n = 73$) and meat shops ($n = 77$) located in urban areas of the Durg and Raipur districts of Chhattisgarh, India. The samples were collected using sticky tapes (~15 cm long) placed near fly congregation areas in milk and meat shops. About 30 flies were collected in sample bags from each shop and brought to the laboratory within 1 h under chilled conditions. The trapped flies were then identified as *Musca domestica* by the Department of Veterinary Parasitology, Dau Shri Vasudev Chandrakar Kamdhenu Vishwavidyalaya, Durg on the basis of macroscopic and microscopic features.

Isolation and identification of *E. coli* from flies

The fly samples were screened for *E. coli* using standard microbiological and molecular methods (Gupta *et al.* 2012). Trapped flies over the sticky tapes were killed by freezing, and individual fly was randomly collected using sterile forceps and placed in a zip-locked plastic bag. The flies in zip-locked plastic bags were crushed using a pestle and inoculated in 5 ml MacConkey's lactose broth (HiMedia, Mumbai, India) followed by incubation at 37°C for 16 h. A loopful of enriched inoculum was then streaked onto eosin methylene blue (EMB) agar (HiMedia) and incubated at 37°C for 16 h. Colonies revealing dark-centred and flat colonies with characteristic metallic sheen on EMB agar were subjected to biochemical tests for identification of *E. coli*. The colonies exhibiting Indole, methyl-red as positive Voges-Proskauer and Citrate utilization tests as negative were confirmed as *E. coli* isolates. Genomic DNA was extracted from pure cultures of *E. coli* isolates using the snap chill method (Khan *et al.* 2021). The extracted DNA was stored at -20°C until further analysis. All isolates were further confirmed as *E. coli* by PCR using 16S rRNA gene-specific primers (Sun *et al.* 2011). The genomic DNA of *E. coli* isolates were also subjected to the detection of fluoroquinolone-resistant *gyrA* and *parC* genes.

Antimicrobial susceptibility test and MAR index

The phenotypic antibiotic resistance of the isolated *E. coli* was determined by the disk diffusion method as per the

guidelines provided by the Clinical and Laboratory Standards Institute (CLSI 2015). The antimicrobials included were ampicillin (AMP, 10 µg), cefotaxime (CTX, 30 µg), ceftazidime (CAZ, 30 µg), cefixime (CFM, 5 µg), aztreonam (AT, 30 µg), ampicillin/cloxacillin (AXE, 10 µg), cephalexin (CN, 30 µg), oxytetracycline (O, 30 µg), ciprofloxacin (CIP, 5 µg) and Gentamicin (GEN, 10 µg) (HiMedia). Results of the antimicrobial sensitivity tests were recorded as sensitive (S), intermediate (I) or resistant (R) and interpretations were carried out as per CLSI standards (CLSI 2015). An *E. coli* isolate resistant to a minimum of three classes of antimicrobials was considered an MDR.

Phenotypic detection of ESBL production in *E. coli*

ESBL producer *E. coli* isolates were detected following the CLSI guidelines. The isolates resistant to cephalosporin discs and sensitive to cephalosporin–clavulanic acid discs with more than 5 mm diameter of zone of growth inhibition were considered as presumptive ESBL producers (CLSI 2015).

Plasmid DNA isolation and purification

Plasmid DNA of presumptive ESBL-producing *E. coli* isolates was isolated from 2 ml of fresh overnight Luria-Bertani broth culture using HipurA® Plasmid DNA Miniprep kit (Cat. No. MB508, Himedia), according to the manufacturer's instructions. The plasmids were stored at −20°C and used for screening of ESBL, *tetA*, *tetB* and *intI1* genes by PCR.

Molecular detection of antibiotic-resistant genes

All the presumptive ESBL-producing and MDR *E. coli* isolates were subjected to the detection of antibiotic-resistant *bla*_{CTX-M}, *bla*_{TEM}, *bla*_{SHV}, *tetA*, *tetB*, *gyrA*, *parC* and integron-associated *intI1* genes by PCR. The specific primers used were synthesized by Eurofins Genomics, India (Table 2). The *bla*_{CTX-M}, *bla*_{TEM} and *bla*_{SHV} genes were targeted for the identification of ESBL, *tetA* and *tetB* genes for tetracyclines, *gyrA* and *parC* genes for fluoroquinolones. Further, *intI1* was targeted for integron1. Resistant *Salmonella* isolates having *bla*_{TEM} and *bla*_{CTX-M} genes preserved in our department (NCBI accession numbers—MN052808 and MN052811) were used as PCR-positive control. The PCR product was separated by gel electrophoresis in 1.5% agarose-containing ethidium bromide in Tris-borate-EDTA buffer and digitally recorded using a gel documentation system (GelDoc™ XR, Bio-Rad, Hyderabad, India).

Hierarchical clustering

Antibiogram results of the 10 antibiotics tested in the present study were recorded in an Excel file and automatically coded as follows: “2” corresponds to resistant, “1” to intermediate and “0” to susceptible. Tested antibiotics and *E. coli* isolates were represented in rows and columns, respectively. Hierarchical clustering of data, based on the three values attributed, was carried out using Hem I software (Heatmap Illustrator, version1.0; Deng et al. 2014), and the data matrix was presented graphically by colouring each antibiotic susceptibility testing result on the basis of measured colour range. The lower limit 0 was coloured

Table 2 Sequences of primers used for detection of antimicrobial resistance and ERIC-PCR genes

Antimicrobial family	Target gene	Primer	Product size (bp)	Reference
Beta-lactams	<i>bla</i> _{SHV}	F ATGCGTTATATTCGCCTGTG R TGCTTTGTTAT CGGGCCAA	747	Paterson et al. (2003)
	<i>bla</i> _{TEM}	F TCGCCGCATACACTATTCTCAGAATGA R ACGCTCACCGGCTCCAGATTTAT	445	Apaka et al. (2010)
	<i>bla</i> _{CTX-M}	F ATGTGCAGYACCAGTAARGTK ATG GC R TGGGTRAARTARGTSACCAGAAYCAGC GG	593	Boyd et al. (2004)
Tetracycline	<i>tetA</i>	F GCTACATCCTGCTTGCCCTTC R CATAGATCGCCGTGAAGAGG	210	Titilawo et al. (2015)
	<i>tetB</i>	F TTGGTTAGGGGCAAGTTTGT R GTAATGGGCAATAACACCG	359	
Fluoroquinolone	<i>gyrA</i>	F CGTCGCTACTTTACGCCATGAACG R ATACCTTGCCGCGACCGGTACGG	586	Dasgupta et al. (2018)
	<i>parC</i>	F TGTATGCGATGTCTGAACGT R CTCAATAGCAGCTCGGAATA	265	
Transposon	<i>int1</i>	F CCTCCGCGACGATGATC R TCCACGCATCGTCAGGC	280	Khoramrooz et al. (2016)
Genotyping	ERIC	F ATG TAA GCT CCT GGG GAT TCA R AAG TAA GTG ACT GGG GTG AGCG	Variable	Ibrahim et al. (2016)

blue, upper limit 2 was coloured red and midpoint value 1 was coloured green.

A hierarchical clustering algorithm involves two main steps that are repeated in order to find the isolates that are most similar. The first step is assigning the antibiogram result values from the same isolate to its own cluster and the second step is merging the two clusters that are closest to each other until only one big cluster develops. Euclidian distance with a complete linkage method was used as a distance matrix.

ERIC-PCR typing

ERIC-PCR was performed using the ERIC-F and ERIC-R primers (Table 2) and cycling conditions were as described by Ibrahim *et al.* (2016). The ERIC-PCR typing analyses were carried out using Paleontological Statistics Software (PAST) Package program version 4.03 (Hammer *et al.* 2001). A cut-off value of 80% similarity was applied to define the clusters. The similarity between the profiles was evaluated with the band-matching Dice coefficient, and dendrograms for each isolate were produced by the Jaccard similarity coefficient and the unweighted pair group method with arithmetic averages (UPGMA) (Ahmed *et al.* 2020). Based on the UPGMA dendrograms, isolates were defined as identical with >97% similarity, closely related with ≥95% similarity and unrelated with <95% similarity.

Acknowledgements

The authors express their gratitude to the Dean, College of Veterinary Science and Animal Husbandry, Durg, Chhattisgarh and Director Research, Dau Shri Vasudev Chandrakar Kamdhenu Vishwavidyalaya, Durg, Chhattisgarh, India for providing necessary facilities to carry out this research work.

Conflict of Interest

The authors declare that they have no conflict of interest.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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